

G11 PHYSICS

Assignment 4: Motion

Due: Monday, 27 April 2026

Total: 34 marks

Scenario. A ball is launched from a rooftop that is $H = 20$ m above the ground. It is launched at an angle of $\theta = 37^\circ$ above the horizontal with an initial speed of $v_0 = 25$ m/s. Take the launch point as the origin, with x pointing horizontally and y pointing vertically upward. Use $g = 10$ m/s², $\sin 37^\circ = 0.6$, and $\cos 37^\circ = 0.8$.

Part A. Kinematics of the Launch

(15 marks)

- (a) Write down the x - and y -components of the initial velocity. Hence state the full position vector

$$\vec{s}(t) = x(t) \hat{x} + y(t) \hat{y}$$

as explicit functions of time.

- (b) Find the time t^* at which the ball reaches its maximum height above the launch point. What is that maximum height H ?
- (c) The ball lands on the ground, which is 20 m *below* the launch point, i.e. $y = -20$ m. Set up the equation for the landing time and solve the resulting quadratic. Clearly state which root you accept and justify why the other root is rejected.
- (d) Calculate the horizontal range (i.e., the horizontal distance from the base of the building to the landing point).

Part B. Velocity, Speed & Acceleration

(11 marks)

- (a) Using the landing time found in Part A(c), find the velocity vector $\vec{v}$ at the moment of impact. Calculate the speed at impact and the angle it makes with the horizontal.
- (b) At what instant(s) during the flight is the speed of the ball equal to its initial speed $v_0 = 25$ m/s? First reason physically, then verify algebraically.
- (c) Write down the acceleration vector $\vec{a}(t)$ throughout the flight. What is its magnitude? Is it ever zero during the flight? Explain briefly.

Part C. Integrals & Trajectory

(8 marks)

- (a) Evaluate the definite integral

$$\int_0^{t^*} v_y(t) dt,$$

where t^* is the time to maximum height found in Part A(b). Interpret the result both geometrically (in terms of area under the v - t graph) and physically (what quantity does it represent?).

- (b) Eliminate t from the parametric equations $x(t)$ and $y(t)$ to obtain the trajectory equation $y = f(x)$. Show that it describes a downward-opening parabola.

Show all working clearly. Marks are awarded for method, not only the final answer. Diagrams are encouraged wherever they aid explanation.